

# Emergent altermagnetic correlations and $d_{xy}$ pairing in the checkerboard Hubbard model

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We study the half-filled Hubbard model on the checkerboard lattice by constrained-path quantum Monte Carlo. The Hamiltonian is spin independent, so the two magnetic sublattices are related by a fourfold rotation rather than by a translation, and any spin splitting of the momentum distribution must be generated by the interaction. We find compensated spin-split correlations above a finite interaction strength whose momentum-space polarization is odd under  $C_4$  and even under the diagonal mirror at all 126 interacting parameter sets studied, which identifies the  $d_{xy}$  channel. A control calculation at vanishing lattice anisotropy, where the model reduces to the square lattice and the splitting is forbidden by symmetry, fixes the statistical floor of the order parameter at  $6.3(2) \times 10^{-4}$  per site, two orders of magnitude below the measured signal. The interaction-induced pairing vertex vanishes identically in the noninteracting limit and is largest in the  $d_{xy}$  channel above an anisotropy  $\delta_c \simeq 0.19$  whose value converges with system size. Magnetism and pairing therefore select the same representation, which distinguishes this state from the  $d_{x^2-y^2}$  pairing reported for the checkerboard lattice in the regime where the two models coincide. Finite-size analysis up to  $16 \times 16$  sites places the extrapolated staggered moment below the resolution of the calculation, so the correlations remain short ranged on the lattices studied.

## I. INTRODUCTION

Altermagnets combine a vanishing net moment with a spin-split band structure, a combination long assumed to require either ferromagnetism or spin-orbit coupling [1, 2]. The splitting follows from the crystal symmetry: the two opposite-spin sublattices are connected by a rotation rather than by a translation or an inversion, so the spin degeneracy that a conventional antiferromagnet enforces is absent.

Most microscopic models of this state build the splitting into the Hamiltonian, usually through a spin-dependent hopping. The splitting is then present already at zero interaction and the question of whether interactions alone can produce it is left open. Two recent studies of the Lieb lattice state the gap directly, noting that unbiased studies establishing an altermagnet as the ground state of an interacting Hamiltonian are lacking [3, 4]. Both use methods with a controlled range of validity, unrestricted Hartree-Fock with exact diagonalization on small clusters in the first case and a functional renormalization group treatment of the weak-coupling limit in the second.

The checkerboard lattice offers a route to the same physics without putting it in by hand. It is a square lattice with diagonal bonds on alternating plaquettes, so the two sublattices are exchanged by a fourfold rotation about a plaquette center but by no translation. The hopping is spin independent, and the interaction is a uniform on-site Hubbard  $U$ . Any spin splitting therefore appears only once magnetic correlations develop, which is the emergent route rather than the built-in one.

We report constrained-path quantum Monte Carlo (CPQMC) calculations for this model at half filling on lattices of  $8 \times 8$  up to  $16 \times 16$  sites. Three results follow. The momentum-resolved spin polarization is

nonzero above a finite  $U$  and has  $d_{xy}$  symmetry at every interacting parameter set we examined. A null calculation at vanishing anisotropy, where symmetry forbids the signal, places the statistical floor two orders of magnitude below the measured values. The interaction-induced pairing vertex is largest in the  $d_{xy}$  channel over the same anisotropy range, so the magnetic and pairing instabilities lie in the same representation.

## II. MODEL AND METHOD

We consider

$$H = \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1)$$

with nearest-neighbor hopping  $t_{ij} = t_0$  and diagonal hopping  $t_1 + t_2$  along  $(1, 1)$  and  $t_1 - t_2$  along  $(1, -1)$  on one sublattice, the two exchanged on the other. Energies are in units of  $|t_0|$ , and we set  $t_0 = -1$  and  $t_1 = 0.3$  throughout. The anisotropy is parametrized by  $\delta$  through  $t_2 = -\delta$ , so that  $\delta = 0$  restores the plain square lattice with uniform second-neighbor hopping and both sublattices become equivalent. All calculations are at half filling.

The hopping in Eq. (1) is spin independent by construction, so  $n_\uparrow(\mathbf{k}) = n_\downarrow(\mathbf{k})$  at  $U = 0$  for every  $\delta$ .

We use CPQMC with a Trotter step  $\Delta\tau = 0.01$ , a projection length  $\beta = 32$ , and  $N_w = 1000$  walkers. The constrained-path approximation introduces a bias controlled by the trial wave function, so we do not describe these results as exact. Doubling the projection length from  $\beta = 32$  to  $\beta = 48$  changes the order parameter by between 0.1% and 4.1% with no systematic drift, which we take as convergence in  $\beta$ .

The order parameter is the momentum-space spin polarization

$$\Delta_{\text{tot}} = \sum_{\mathbf{k}} |n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})|, \quad (2)$$

following Ref. [6]. The absolute value inside the sum makes  $\Delta_{\text{tot}}$  sensitive to a compensated splitting, for which the unsigned sum  $\sum_{\mathbf{k}} [n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})]$  vanishes. We quote  $\Delta_{\text{tot}}/N$  so that different lattice sizes can be compared, and we note that Eq. (2) as written in Ref. [6] is a bare sum.

Because  $H$  commutes with the total spin, the polarization must be induced by a symmetry-breaking field. We add a staggered field  $h$  that couples with opposite sign to the two sublattices, which makes the trial state symmetry broken, and we report the behavior as a function of  $h$ . The constrained path is defined by the trial state, so this field biases the walkers toward the broken state, and the control described below is what establishes that the measured signal is not an artifact of that bias.

To classify the symmetry of  $\Delta n(\mathbf{k}) = n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})$  we use two dimensionless ratios,

$$A_{\text{odd}} = \frac{\sum_{\mathbf{k}} |\Delta n(\mathbf{k}) + \Delta n(R_{90}\mathbf{k})|}{\sum_{\mathbf{k}} |\Delta n(\mathbf{k})|}, \quad (3)$$

$$M_{\text{odd}} = \frac{\sum_{\mathbf{k}} |\Delta n(\mathbf{k}) + \Delta n(M\mathbf{k})|}{\sum_{\mathbf{k}} |\Delta n(\mathbf{k})|}, \quad (4)$$

where  $R_{90}$  is the fourfold rotation and  $M$  the diagonal mirror  $(k_x, k_y) \rightarrow (k_y, k_x)$ . A state odd under  $C_4$  gives  $A_{\text{odd}} = 0$  and even gives  $A_{\text{odd}} = 2$ . A  $d_{x^2-y^2}$  state gives  $M_{\text{odd}} = 0$  and a  $d_{xy}$  state gives  $M_{\text{odd}} = 2$ . These ratios use only the symmetry of the data, so unlike a projection onto a single lattice harmonic they are not sensitive to how the weight is distributed among higher harmonics of the same symmetry.

### III. EMERGENT ALTERMAGNETIC CORRELATIONS

$\Delta_{\text{tot}}$  vanishes identically at  $U = 0$  and becomes finite for  $U > 0$  at every anisotropy studied, reaching  $\Delta_{\text{tot}}/N$  between  $2.2 \times 10^{-3}$  and  $9.2 \times 10^{-2}$  over the interacting parameter sets. The splitting is compensated, so the state carries momentum-resolved spin polarization with no net moment.

The symmetry classification is unambiguous. Across all 126 interacting parameter sets at  $L = 8, 10$ , and  $12$  we obtain  $A_{\text{odd}}$  between  $0.004$  and  $0.263$  and  $M_{\text{odd}}$  between  $1.24$  and  $2.00$ . Restricted to  $L = 12$  the ranges are  $0.008$  to  $0.166$  and  $1.54$  to  $2.00$ . The polarization is therefore odd under  $C_4$  and even under the diagonal mirror at every parameter set, which is the  $d_{xy}$  channel. No parameter set is classified as  $d_{x^2-y^2}$ .

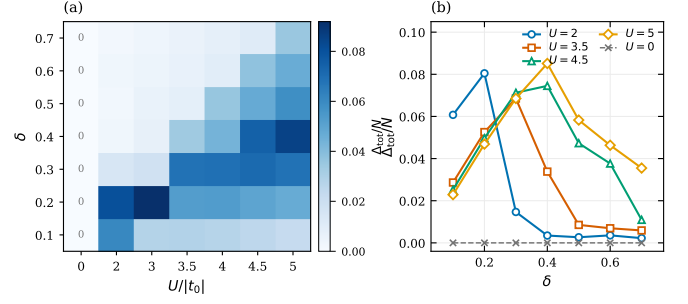


FIG. 1. Momentum-space spin polarization  $\Delta_{\text{tot}}/N$  over the  $(U, \delta)$  plane at half filling,  $L = 12$ . The  $U = 0$  column is identically zero at every anisotropy, which is required because the hopping in Eq. (1) is spin independent. The signal appears only once the interaction is turned on, which is the emergent route rather than the built-in one.

### IV. NULL CONTROL AT VANISHING ANISOTROPY

Equation (2) sums absolute values, so statistical noise contributes with a definite sign and  $\Delta_{\text{tot}}$  has a positive floor that does not vanish when the physics requires it to. Establishing the scale of that floor requires a calculation in which the signal is forbidden but the sampling is unchanged.

The  $U = 0$  column does not serve this purpose. There the Green function is obtained without auxiliary fields, so the zero is a matter of arithmetic rather than of statistics and it tests nothing about the sampling.

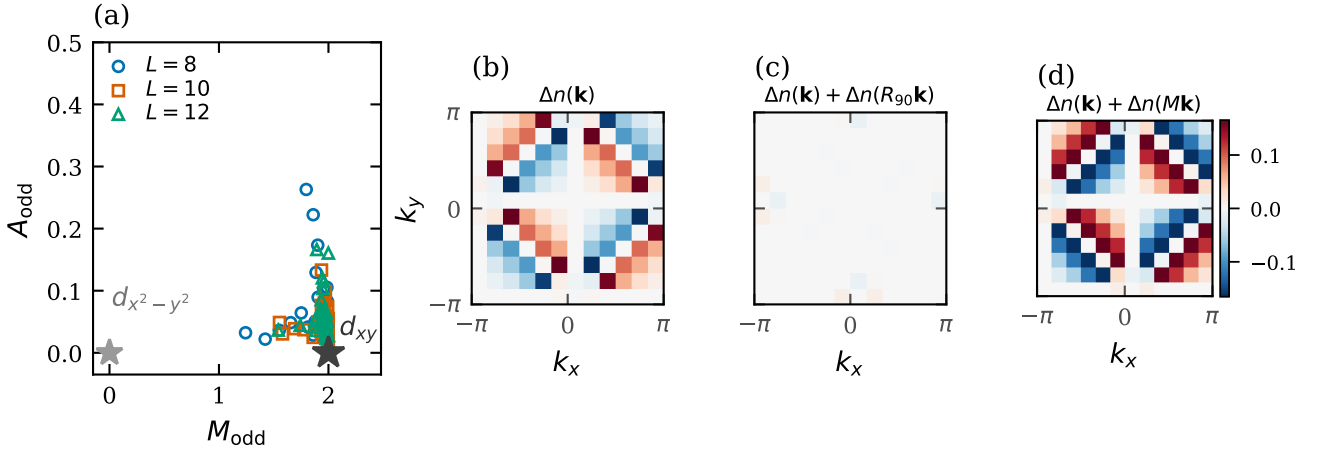
At  $\delta = 0$  both diagonal bonds are equal on every site, the two sublattices become equivalent, and the model reduces to the square lattice with uniform second-neighbor hopping. Its Néel sublattices are related by a translation, so  $n_{\uparrow}(\mathbf{k}) = n_{\downarrow}(\mathbf{k})$  identically and  $\Delta_{\text{tot}}$  must vanish by symmetry at any  $U$ . Whatever is measured there is the statistical floor at production settings.

We find  $\Delta_{\text{tot}}/N = 6.3(2) \times 10^{-4}$ , averaged over  $U = 2, 3, 3.5, 4, 4.5$ , and  $5$  at  $L = 12$ . At  $\delta = 0.3$  the measured signal ranges from  $1.47 \times 10^{-2}$  at  $U = 2$  to  $7.13 \times 10^{-2}$  at  $U = 4.5$ , which is between 23 and 114 times this floor.

### V. FINITE-SIZE BEHAVIOR OF THE MAGNETIC CORRELATIONS

The polarization of Sec. III is measured on finite lattices, and a nonzero value there does not by itself establish order in the thermodynamic limit. We therefore examine two quantities that distinguish a genuine order parameter from a finite-size remnant.

The first is the equal-time spin structure factor at the ordering wave vector,  $S(\pi, \pi)/N$ . Across  $L = 8, 10$ , and  $12$  and for densities from  $n = 0.5$  to  $n = 1$  at  $U$  up to  $8$ , the ratio between successive sizes falls between  $0.39$  and  $0.54$ , against  $0.44$  for a pure  $1/N$  decay. A structure factor that scales as  $1/N$  carries no long-range order, so this



$$L = 12, U = 4, \delta = 0.4$$

FIG. 2. Symmetry classification of  $\Delta n(\mathbf{k})$  for all 126 interacting parameter sets at  $L = 8, 10$  and  $12$ . The ratios  $A_{\text{odd}}$  of Eq. (3) and  $M_{\text{odd}}$  of Eq. (4) place every cell at  $A_{\text{odd}} \simeq 0$  and  $M_{\text{odd}} \simeq 2$ , which is odd under  $C_4$  and even under the diagonal mirror and identifies the  $d_{xy}$  channel. No cell is classified as  $d_{x^2-y^2}$ .

half filling.  $L = 14$  forms a curve only for  $\delta \leq 0.3$ ; its single  $\delta = 0.4$  cell is omitted.

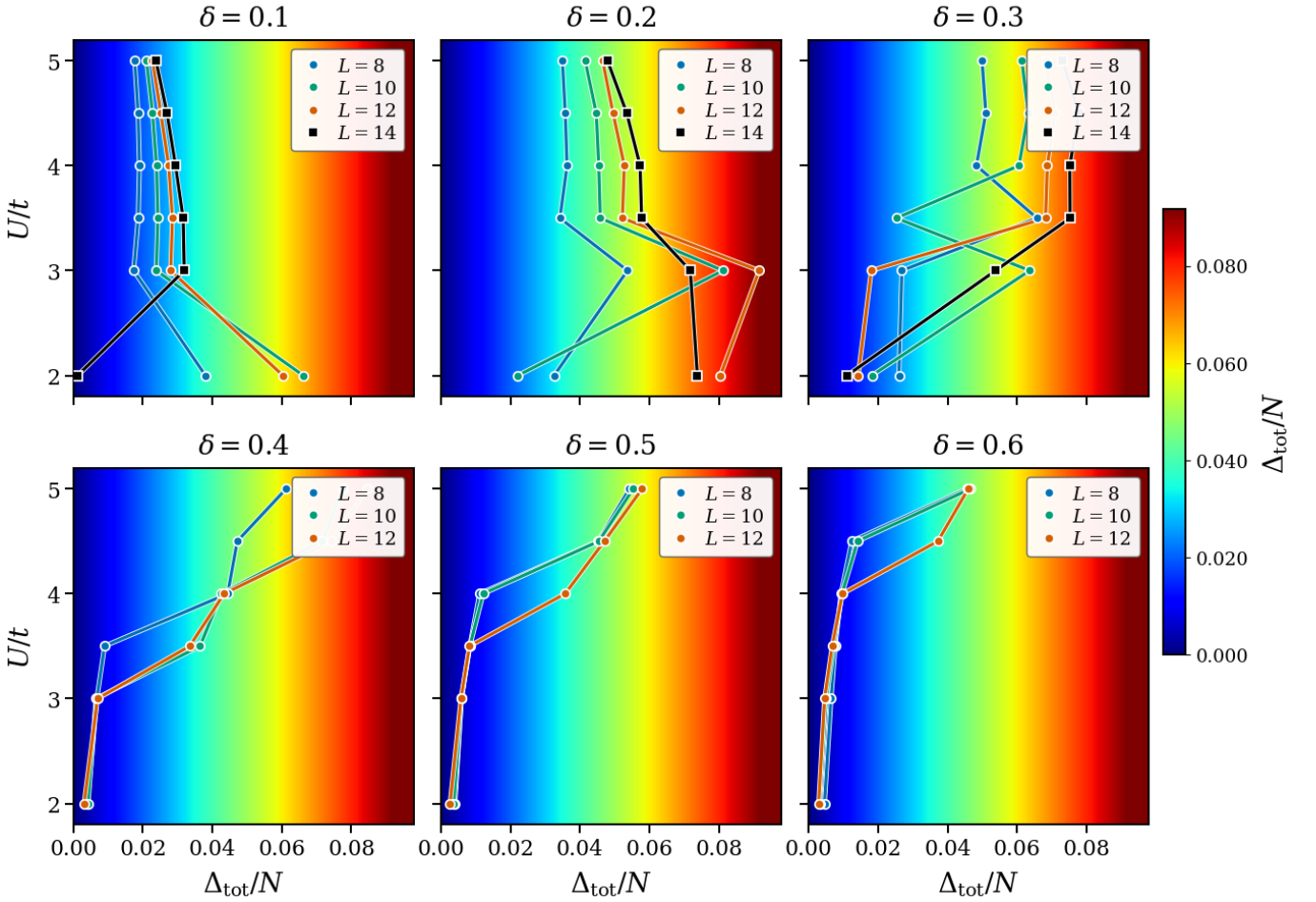


FIG. 3.  $\Delta_{\text{tot}}/N$  against  $U$  at six anisotropies from  $\delta = 0.1$  to  $0.6$ , for  $L = 8, 10, 12$  and  $14$ . The background shades the value of  $\Delta_{\text{tot}}/N$  on the same scale as the heat maps so that the two presentations can be read against each other.  $L = 14$  forms a curve only for  $\delta \leq 0.3$ .

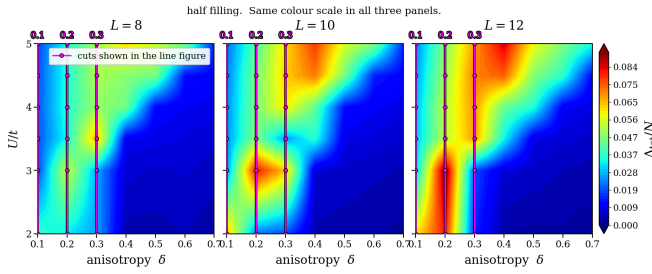


FIG. 4.  $\Delta_{\text{tot}}/N$  over the  $(U, \delta)$  plane with the cuts used in the preceding figure drawn on the map,  $L = 12$ .

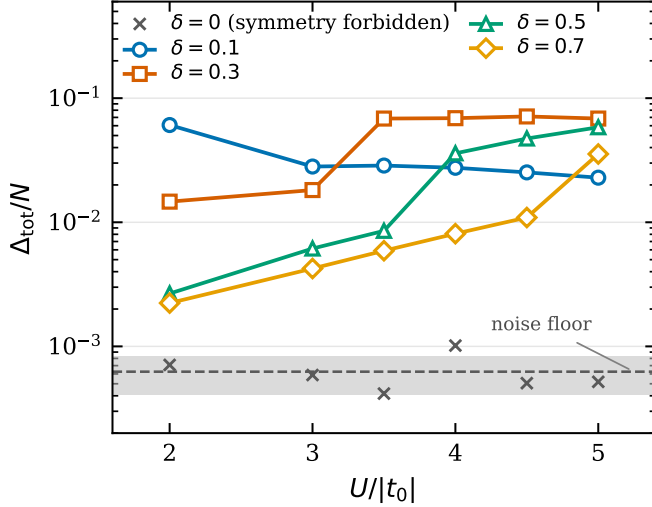


FIG. 5. Null control at vanishing anisotropy, logarithmic axis. At  $\delta = 0$  the two sublattices are related by a translation and  $\Delta_{\text{tot}}$  must vanish by symmetry at any  $U$ , so what is measured there is the statistical floor of the estimator at production settings. The floor is  $6.3(2) \times 10^{-4}$  per site and the measured signal at  $\delta = 0.3$  sits between 23 and 114 times above it.

alone leaves the correlations short ranged on the lattices reached here.

The second is the extrapolation of the polarization to vanishing field. The  $h \rightarrow 0$  intercept of  $\Delta_{\text{tot}}/N$  is nonzero at each individual size and decreases monotonically with  $L$ , taking the values 0.03269, 0.02363, 0.01845, and 0.01714 at  $L = 8, 10, 12$ , and  $14$ . On those four points alone the extrapolation is ambiguous. A fit in  $1/L^2$  gives  $\chi^2/\text{dof} = 2.1$  against 7.6 for a fit in  $1/L$ , which favors a finite intercept and therefore order.

The  $L = 16$  calculation resolves the ambiguity. The two fits predict 0.01325 in  $1/L$  and 0.01457 in  $1/L^2$ , and the measured value is  $0.00734 \pm 0.00109$ . The drop from  $L = 14$  is a factor of 0.43 where a  $1/L$  law predicts 0.88, and including the point raises the  $1/L^2$   $\chi^2/\text{dof}$  from 2.1 to 11.9. The intercept is a finite-size effect and the magnetic correlations remain short ranged.

This is a statement about the correlation length, not about the symmetry of the polarization. The classification of Sec. III rests on ratios of sums over the Brillouin

zone at a fixed lattice size, so it is unaffected by the scaling of the correlation length.

## VI. PAIRING IN THE SAME REPRESENTATION

We compute the pair-field susceptibility in four channels, on-site  $s$ , extended  $s$ ,  $d_{x^2-y^2}$ , and  $d_{xy}$ , and separate the interaction-induced vertex from the uncorrelated part. The vertex is the meaningful measure of whether the interaction builds pairing, since the full correlator is large and positive already for free electrons.

Three features stand out. The vertex is zero in all four channels at  $U = 0$ , so everything that follows is interaction generated. The on-site  $s$  vertex is negative throughout, as a repulsive  $U$  requires. The  $d_{xy}$  vertex is the largest of the four for  $\delta \geq 0.3$  at every  $U$  studied, and it grows monotonically with  $U$ , from 3.347(8) at  $U = 2$  to 5.952(22) at  $U = 4.5$ , in both cases peaking at  $\delta = 0.4$ .

The leading channel changes with anisotropy. For  $\delta \leq 0.2$  the extended  $s$  channel is largest, and for  $\delta \geq 0.3$  the  $d_{xy}$  channel is. This crossover falls in the same range of  $\delta$  where the slope of  $\Delta_{\text{tot}}$  with respect to  $U$  changes sign, so two quantities measured independently change character together.

The magnetic and pairing instabilities therefore select the same representation. The polarization is  $d_{xy}$  at all 126 parameter sets, and the pairing vertex is largest in  $d_{xy}$  over the anisotropy range where that classification holds most strongly.

## VII. SIZE DEPENDENCE OF THE PAIRING VERTEX

The pairing vertex of Sec. VI was computed at  $L = 8, 10$ , and  $12$  under identical settings, which allows the anisotropy of the channel crossover to be followed with size.

The crossover anisotropy  $\delta_c$ , defined as the value at which the  $d_{xy}$  vertex overtakes the extended  $s$  vertex, is nearly independent of  $U$  and becomes more so as the lattice grows. Its spread across the six interaction strengths studied is 0.148 at  $L = 8$ , 0.049 at  $L = 10$ , and 0.028 at  $L = 12$ , and the mean value moves from 0.235 to 0.197 to 0.192. A crossover that is set by the interaction strength would not converge in this way, so we read  $\delta_c \approx 0.19$  as a property of the lattice geometry.

The  $d_{xy}$  vertex grows with system size at every interaction strength, by a factor between 2.18 and 2.38 from  $L = 8$  to  $L = 12$ . The  $d_{x^2-y^2}$  vertex is positive in all 126 cells at all three sizes, so that channel is never repulsive even where it is subdominant. The optimal anisotropy for  $d_{xy}$  is  $\delta = 0.4$  at essentially every size and interaction strength.

Magnetism and pairing move together rather than merely coexisting. With  $\delta$  and  $U$  both partialled out,

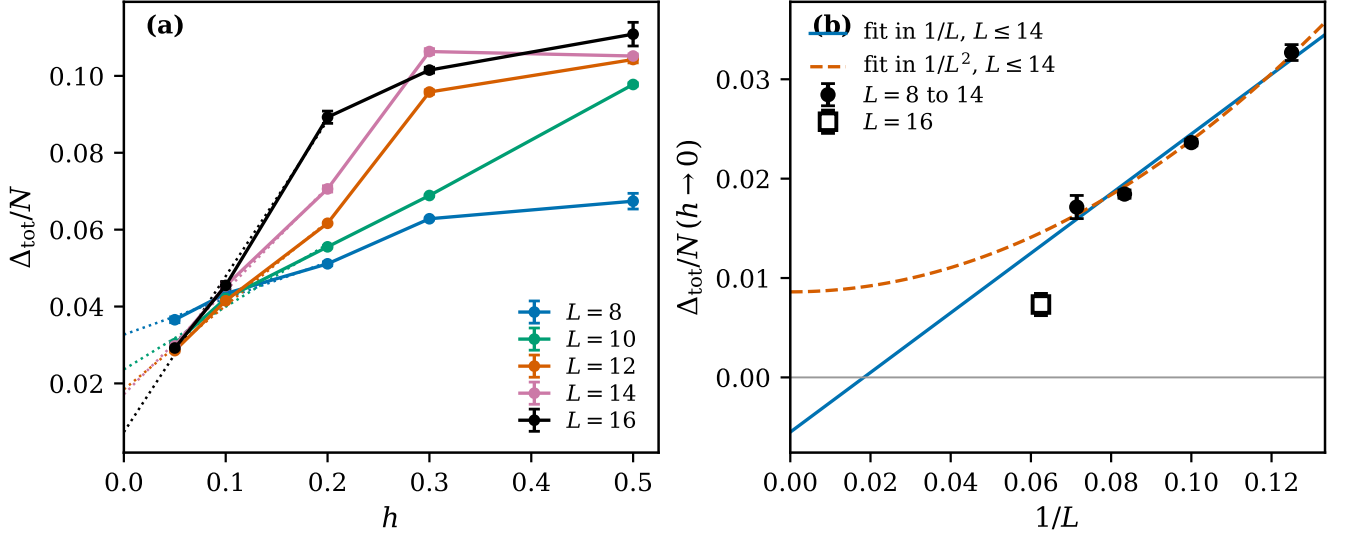


FIG. 6. (a)  $\Delta_{\text{tot}}/N$  against the symmetry-breaking field  $h$  for  $L = 8$  to  $16$  at  $U = 4$  and  $\delta = 0.3$ , with the straight-line fits over the three smallest fields that define the  $h \rightarrow 0$  intercept. (b) those intercepts against  $1/L$ . The two curves are fits to the  $L \leq 14$  points alone, linear in  $1/L$  and in  $1/L^2$ , extended to  $L = 16$ . They predict  $0.01325$  and  $0.01457$  there, and the measured value is  $0.00733 \pm 0.00106$ . The drop from  $L = 14$  is a factor of  $0.43$  where a  $1/L$  law gives  $0.88$ , so the nonzero intercept seen at each individual size is a finite-size effect.

half filling,  $L = 8$  to  $14$ . The  $\delta = 0.1$  peak is gone at  $L = 14$ .

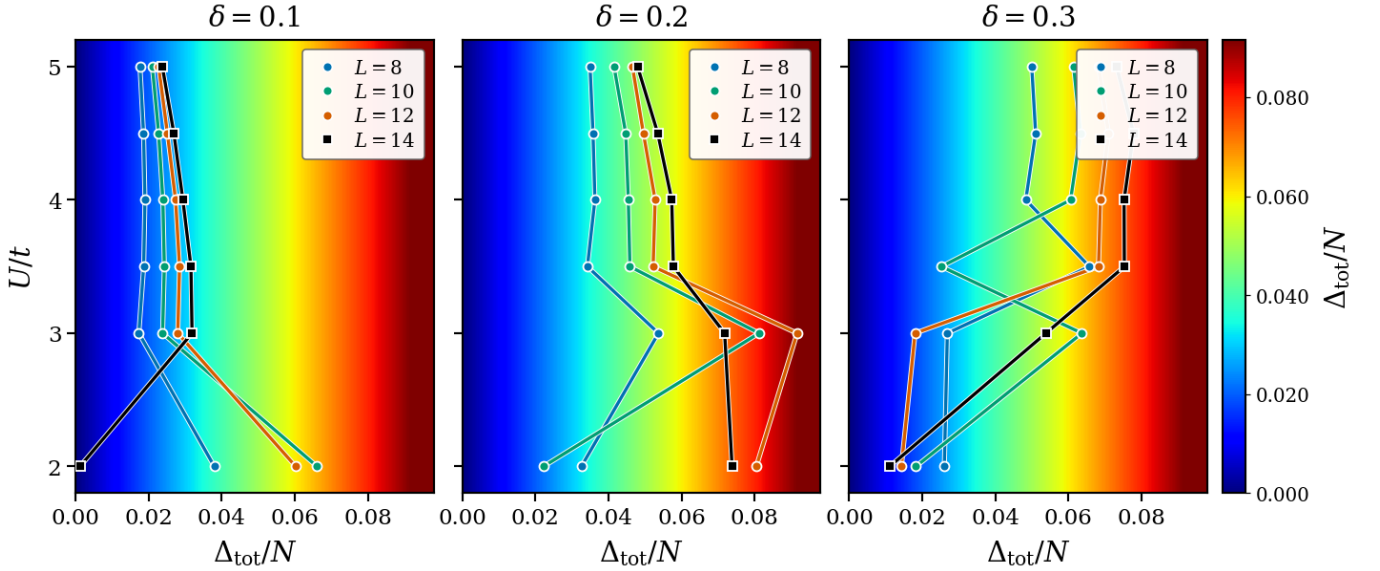


FIG. 7. Behavior at small  $U$  for  $\delta = 0.1, 0.2$  and  $0.3$  across  $L = 8$  to  $L = 14$ . The feature near  $U = 2$  does not survive to  $L = 14$  at  $\delta = 0.1$ , where  $\Delta_{\text{tot}}/N$  falls to about twice the  $\delta = 0$  floor and the correlation with  $\sin k_x \sin k_y$  is lost. At  $\delta = 0.3$  the decay with size is steady. The  $\delta = 0.2$  column persists at every size.

the correlation between the pairing vertex and  $\Delta_{\text{tot}}$  is  $+0.57$  in the  $d_{xy}$  channel and  $-0.51$  in the  $d_{x^2-y^2}$  channel. The two channels respond to the magnetic signal with opposite sign at fixed anisotropy and fixed interaction.

## VIII. ROBUSTNESS AND COMPARISON

At half filling with periodic boundaries the noninteracting gap of Eq. (1) is zero at every  $L$  and  $\delta$  we examined, so the occupied set is fixed by the trial state rather than by the spectrum. We repeated the calcula-

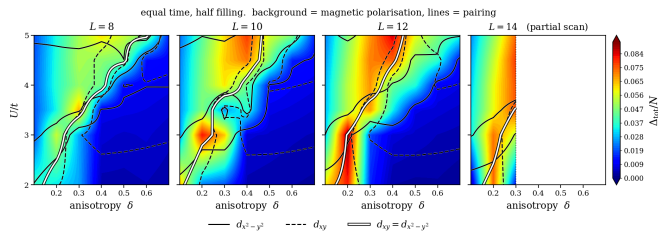


FIG. 8. The four lattice sizes overlaid, so that the size dependence of  $\Delta_{\text{tot}}/N$  can be read at fixed  $U$  and  $\delta$ .

tion with antiperiodic boundaries, which shift the momentum mesh off the high-symmetry points and open a gap, at otherwise identical settings.

The sign of  $\partial\Delta_{\text{tot}}/\partial U$  agrees between the two boundary conditions at  $\delta = 0.1$ , where both are negative, and at  $\delta = 0.5$ , where both are positive, with magnitudes matching to within a few percent despite the different meshes. At  $\delta = 0.3$  the two disagree, and the antiperiodic value is consistent with zero. We read  $\delta \approx 0.3$  as the crossover where the slope passes through zero, at which point the sign is most sensitive to the mesh, and we therefore do not quote a slope at that anisotropy.

The closest published work is a determinant quantum Monte Carlo study of the checkerboard Hubbard model that reports  $d$ -wave and  $d+is$  pairing as the dominant channels [5]. The two Hamiltonians are not the same. That model keeps one diagonal per sublattice and omits the other, whereas Eq. (1) keeps both, and the two coincide only where our second diagonal vanishes, at  $\delta = t_1 = 0.3$ . We verified that the single-particle spectra agree there to within  $4 \times 10^{-15}$ .

At that point, and at half filling, we evaluated the vertex in the basis of Ref. [5] alongside our own. Within their basis the  $s$ -wave channel is the larger of the two at  $U = 3, 4$ , and  $4.5$ . Including  $d_{xy}$ , which their basis does not contain, gives a  $d_{xy}$  vertex larger than either. The comparison is restricted, since Ref. [5] works at density  $n = 0.9$  and temperature  $T/t = 1/6$  while our calculations are at half filling in the ground state, and under the particle-hole transformation that relates the two signs of the second-neighbor hopping their density maps to  $n = 1.1$ . The statement is that the dominant channel we find lies outside the basis used there, not that the two calculations disagree.

## IX. CONCLUSION

The half-filled checkerboard Hubbard model supports compensated spin-split correlations that are absent at  $U = 0$  and are  $d_{xy}$  in symmetry at every parameter set examined. A control at vanishing anisotropy, where the splitting is forbidden, places the statistical floor two orders of magnitude below the signal. The interaction-induced pairing vertex is largest in the same  $d_{xy}$  channel, and the anisotropy at which the pairing channel changes

coincides with the anisotropy at which the magnetic response changes sign.

The mechanism uses no spin-dependent hopping and no spin-orbit coupling. It requires only a lattice whose magnetic sublattices are related by a rotation and a uniform Hubbard repulsion, which suggests that the same construction should apply to other lattices with the same sublattice relation. The correlations remain short ranged on the lattices reached here, and the constrained-path bias remains.

## Appendix A: Trial wave function and the selection of the constrained path

The constrained-path approximation replaces the sign problem with a bias fixed by the trial wave function, so the trial is part of the method rather than an implementation detail. We take the trial from an unrestricted Hartree-Fock solution of Eq. (1) at the same  $U$ ,  $\delta$ , and filling, and we select among the solutions reached from different starting densities by the variational energy

$$E_{\text{HF}} = \text{Tr}(K\rho_{\uparrow}) + \text{Tr}(K\rho_{\downarrow}) + U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle, \quad (\text{A1})$$

with  $\rho_{\sigma} = V_{\sigma} V_{\sigma}^{\dagger}$  built from the occupied columns of the mean-field eigenvectors and  $K$  the one-body matrix of Eq. (1).

Equation (A1) is not interchangeable with the sum of mean-field eigenvalues. Each eigenvalue already contains the mean field of the opposite spin, so that sum counts the interaction twice and orders competing solutions differently from the energy. The two criteria can select different states, and because the constrained path is fixed by whichever state is chosen, the selection propagates into every measured quantity.

The size of the effect is not small. Selecting by the eigenvalue sum instead of Eq. (A1) changes the trial in a substantial fraction of parameter sets, and at half filling the state chosen that way carries the larger staggered moment in every case where the two differ. We therefore report results obtained with Eq. (A1) throughout.

The identity  $E_{\text{HF}} = \sum_p \varepsilon_p - U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle$  holds only at exact self consistency and should not be substituted for Eq. (A1). A mean-field iteration that mixes densities between steps does not reach that condition for loosely converged solutions, which are precisely the ones whose ranking is at issue.

## Appendix B: Convergence in projection length

All results use a projection length  $\beta = 32$  in units of  $1/|t_0|$ . Doubling it to  $\beta = 48$  at otherwise identical settings changes  $\Delta_{\text{tot}}$  by between 0.1% and 4.1% with no systematic drift in either direction, which we take as convergence.

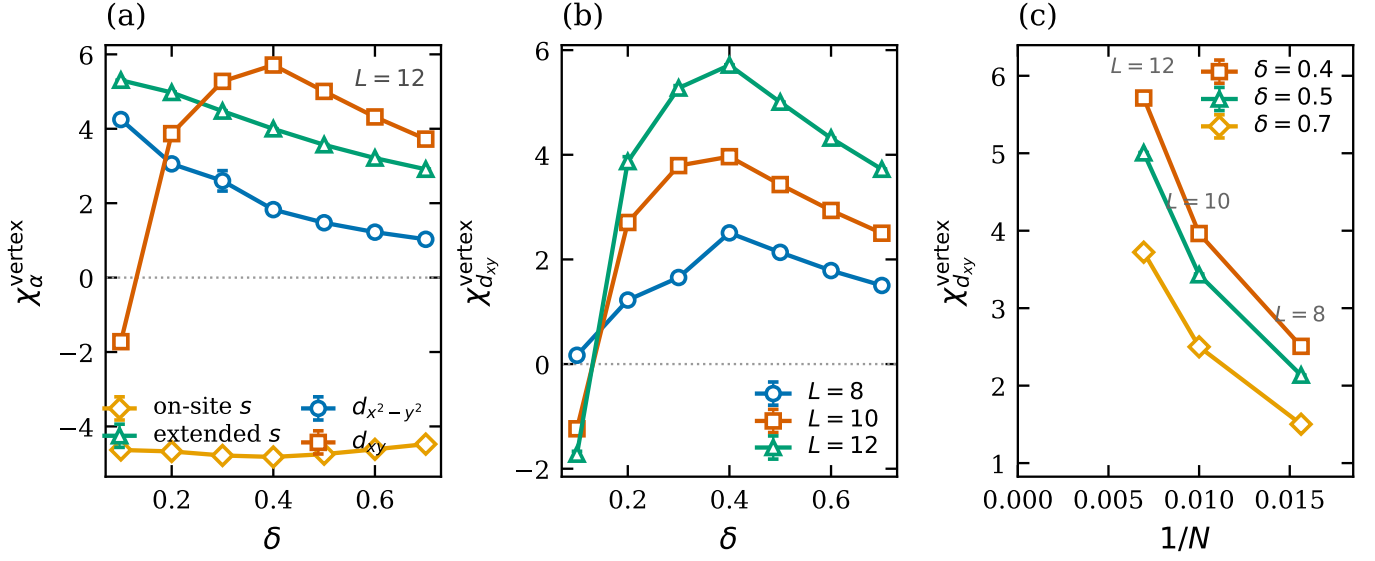


FIG. 9. Time-integrated pairing vertex at half filling. (a) the four channels at  $L = 12$  and  $U = 4$ , (b) the  $d_{xy}$  vertex against  $\delta$  for  $L = 8, 10$  and  $12$ , and (c) the same quantity against  $1/N$ . The vertex is the interaction-induced part of the pair-field correlator and vanishes identically at  $U = 0$  in all four channels.  $d_{xy}$  grows with system size at every interaction strength studied.

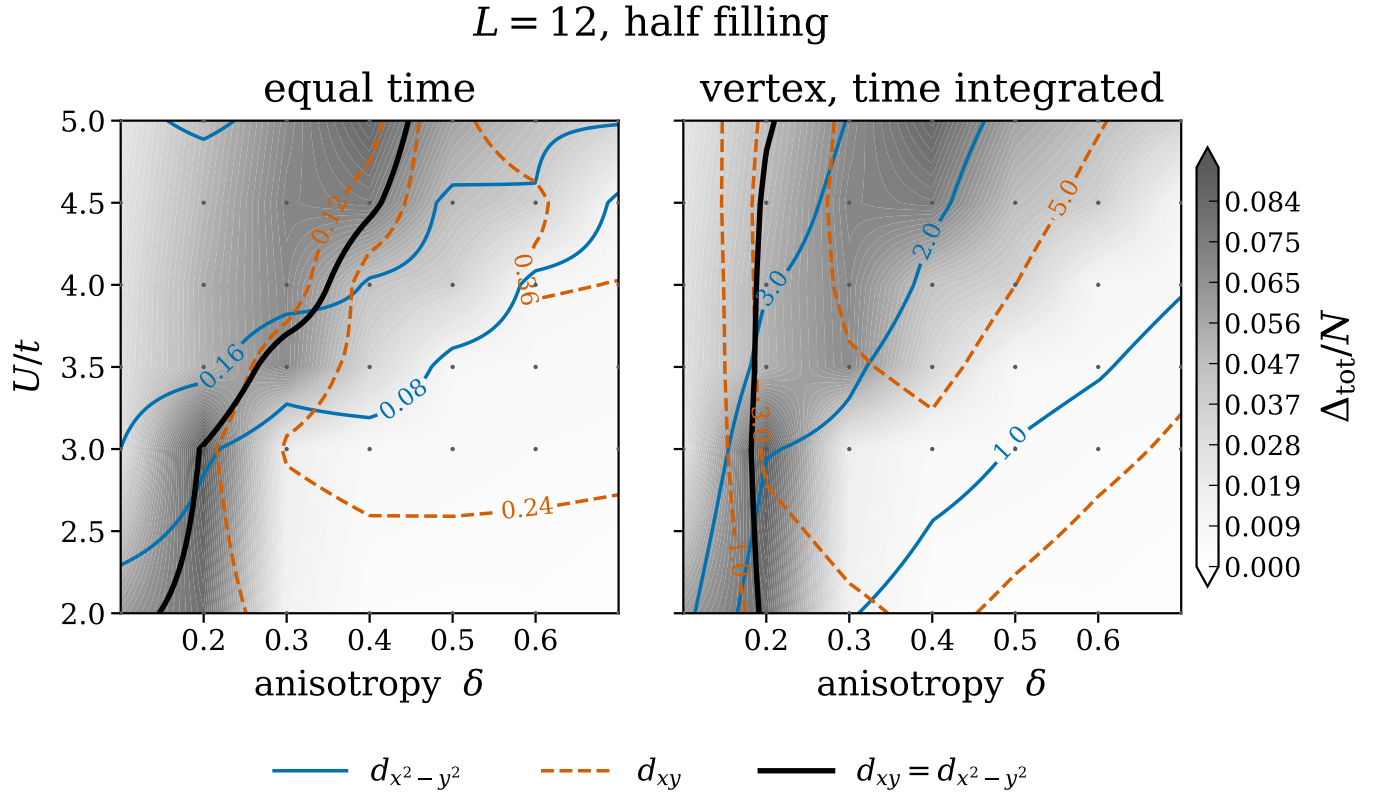


FIG. 10. The two  $d$  channels measured equal time (left) and time integrated (right) at  $L = 12$ . The equal-time correlator carries a large uncorrelated part, so the vertex is the measure used for the claims in the text.

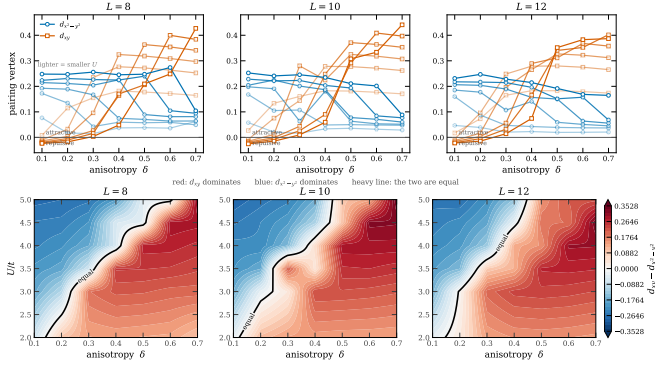


FIG. 11. Both  $d$  channels in the  $(U, \delta)$  plane at  $L = 12$ .  $d_{x^2-y^2}$  is positive everywhere, so it is never repulsive even where it is subdominant, and  $d_{xy}$  is the larger of the two above  $\delta \simeq 0.19$ .

### Appendix C: Boundary conditions

At half filling with periodic boundaries the noninteracting spectrum of Eq. (1) is gapless at every  $L$  and  $\delta$

examined, so the occupied set is fixed by the trial state rather than by the spectrum. Antiperiodic boundaries shift the momentum mesh off the high-symmetry points and open a gap.

The sign of  $\partial\Delta_{\text{tot}}/\partial U$  agrees between the two choices at  $\delta = 0.1$ , where both are negative, and at  $\delta = 0.5$ , where both are positive, with magnitudes matching to within a few percent. At  $\delta = 0.3$  the two disagree and the antiperiodic value is consistent with zero. We read  $\delta \approx 0.3$  as the anisotropy at which the slope passes through zero, where its sign is most sensitive to the mesh, and we do not quote a slope there.

### ACKNOWLEDGMENTS

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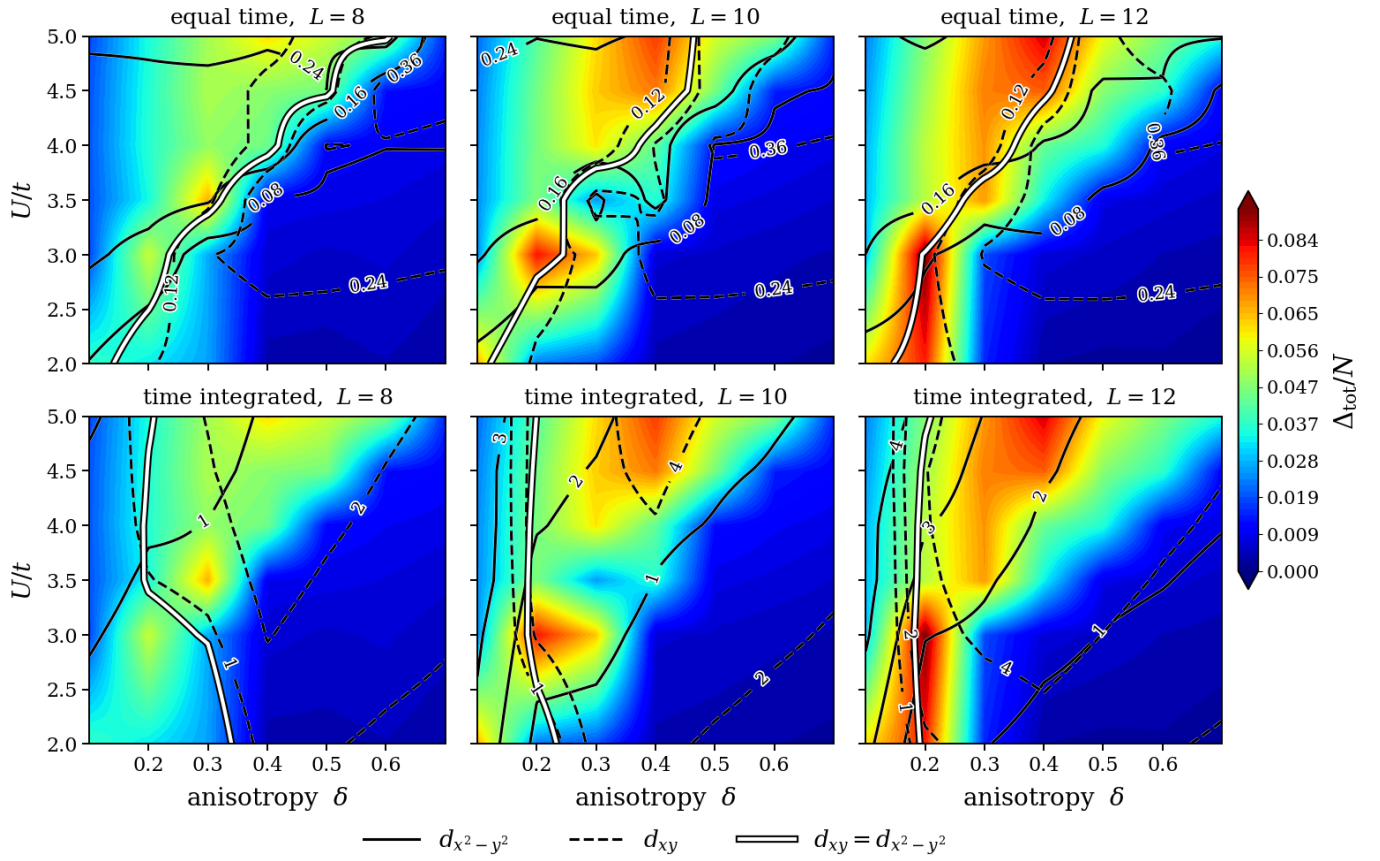


FIG. 12. Equal-time pair correlator (top row) and time-integrated vertex (bottom row) for  $L = 8, 10$  and  $12$ . The channel crossover straightens and its spread across  $U$  narrows along the vertex row, which is the behavior expected of a quantity fixed by the lattice geometry rather than by the coupling.

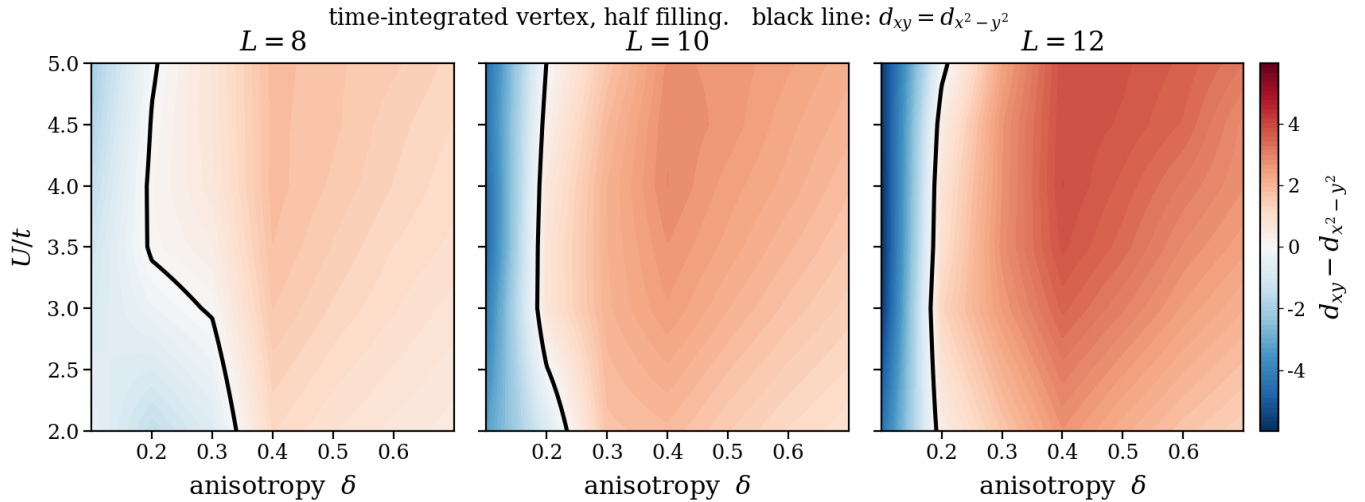


FIG. 13. Difference between the  $d_{xy}$  and  $d_{x^2-y^2}$  vertices over the  $(U, \delta)$  plane at  $L = 8, 10$  and  $12$ . The zero contour is the crossover anisotropy  $\delta_c$ , which is nearly flat in  $U$  and whose spread across the interaction strengths studied falls from 0.148 to 0.049 to 0.028 as the lattice grows.

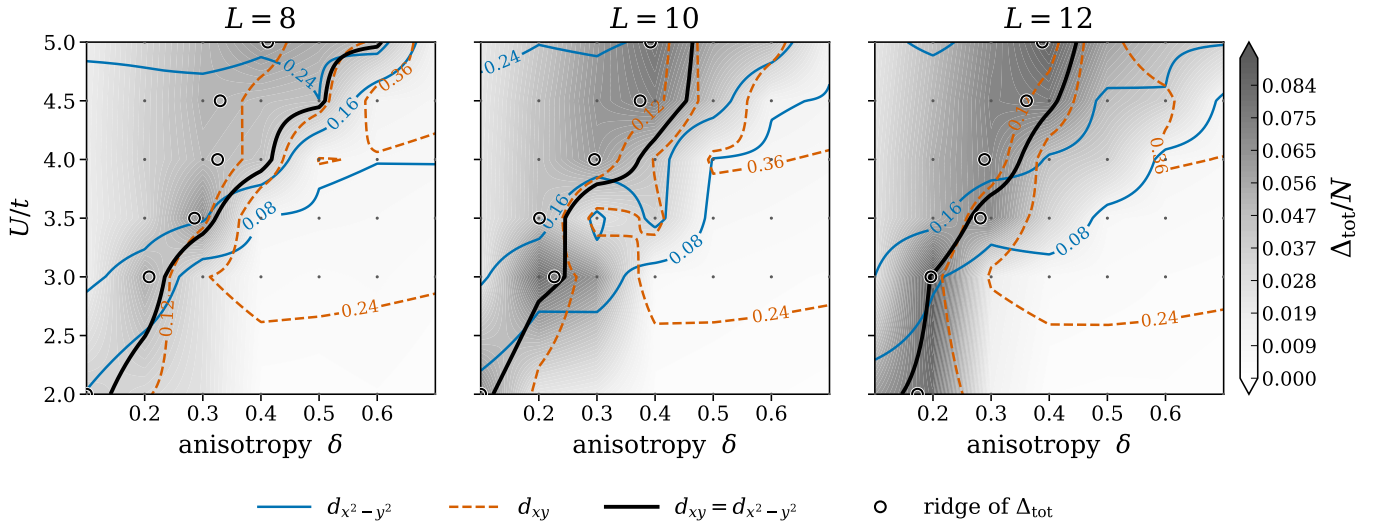


FIG. 14. Contours of the two  $d$  vertices drawn over the measured  $\Delta_{\text{tot}}/N$ . The pairing channel that leads changes across the same region of the plane in which the magnetic signal changes character, so two independently measured quantities turn over together.

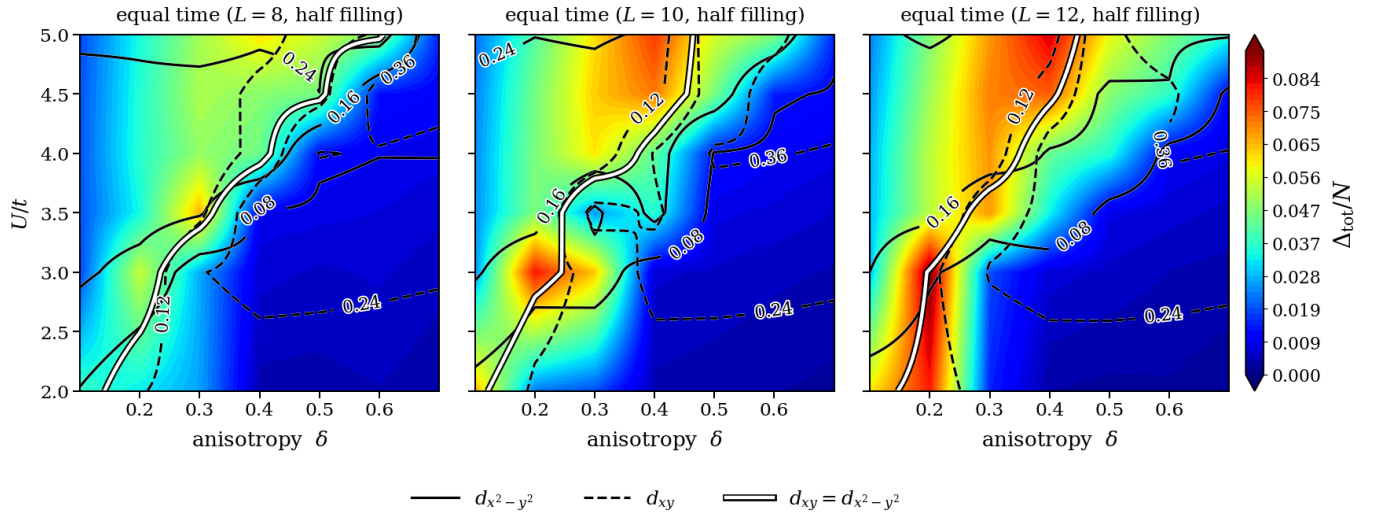


FIG. 15. Pairing contours on the  $\Delta_{\text{tot}}/N$  background for  $L=8$ , 10 and 12, showing that the relation between the two survives the change of lattice size.